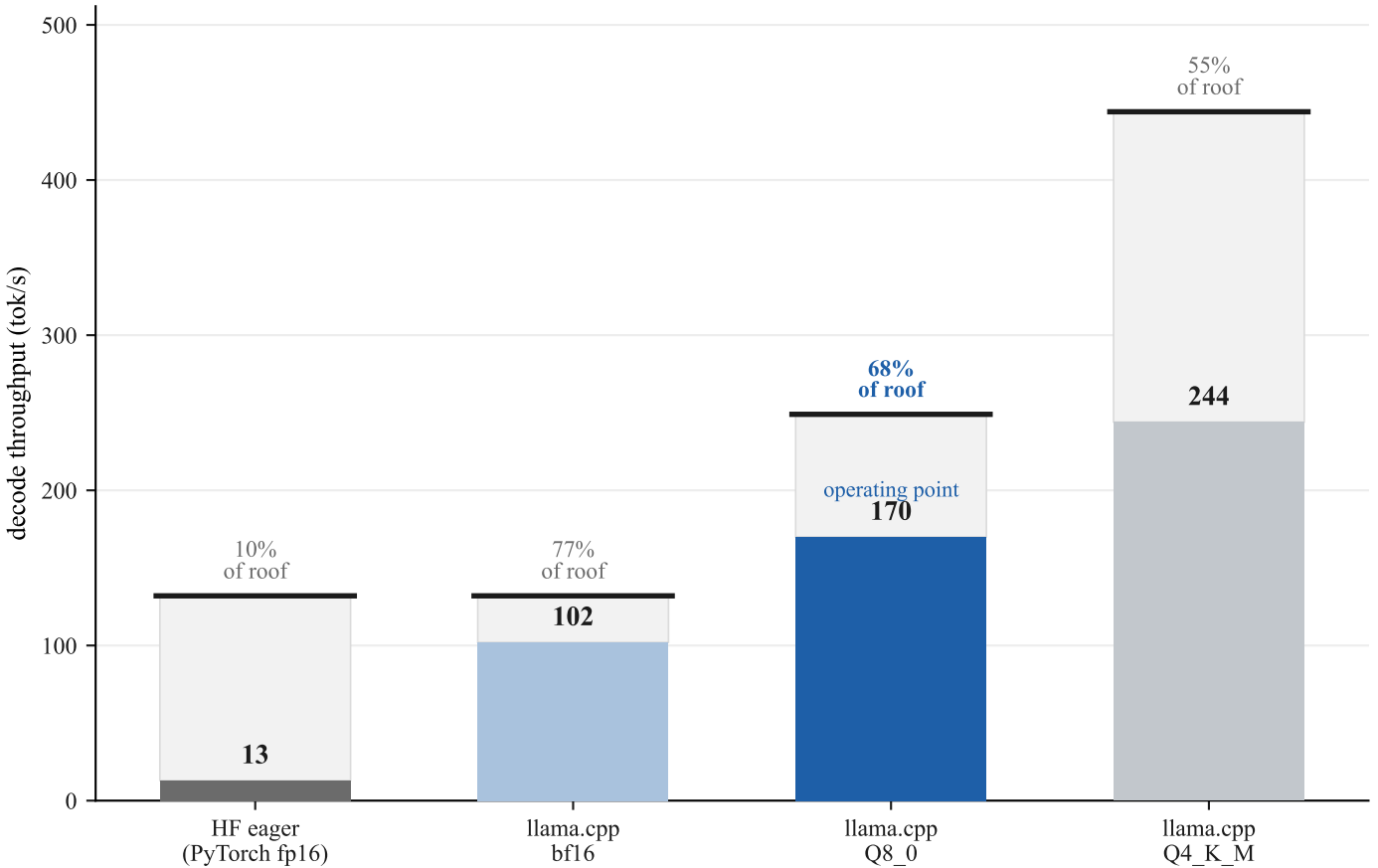
Decode is memory-bandwidth-bound

measured throughput vs the 896 GB/s roof; caps mark each precision's ceiling



eager and llama.cpp bf16 run the same weights at the same roof, so their ~8x gap is pure engine overhead (no CUDA graphs), not the card.
Roof = 896 GB/s / weight bytes per token (bf16 6.77, Q8_0 3.60, Q4_K_M 2.02 GB); a first-order model, KV-cache reads add a few percent.